

SBTF Puerto Rico Mapping Intel



Standby Task Force (SBTF) geolocated highway status and damage data mined from social media and news outlets from October 2–6, 2017 and from Civil Air Patrol / NOAA imagery missions. This mapping is available at www.tiny.cc/sbtf-pr

SBTF is currently working with independent volunteers and GIS Corps to analyse this data and gather information that could be of use to planning and implementing the recovery phase of the operations in PR. Details of this analysis (explaining it's proposed use) and methodology is in the meta data that will be constantly updated and in

<https://docs.google.com/document/d/135kTYzIDqerXRYifHd95HbjS5J1ZJZ9BxOID2nWEsuk/edit?usp=sharing>

To help us track how our data is being used, please provide us with your name, your organization and a brief summary of how you plan to use it to coreteam@standbytaskforce.com

The information provided in this map is intended for humanitarian aid and disaster response, and should be used for these purposes. The data was found using secondary sources and, while attempts were made to find credible sources, cannot be 100% verified.

The Standby Task Force is not responsible for any actions as a result of using the data. Any reference to the map and its content should cite Standby Task Force Inc

i) SBTF-CAP Points by Status-Oct 8 2017-2200UTC

1) Text:

Between October 2nd and October 6th, SBTF and GISCorps volunteers evaluated and geolocated over 1,000 Civil Air Patrol (CAP) oblique aerial photographs taken after Hurricanes Irma and Maria. Highways identified in the images were assessed for status and damage. In this layer, the resulting points are symbolized here according to status; clicking on a

point symbol opens a popup box containing the CAP image and descriptive text.

Status codes:

0 = Open

1 = Partially open

2 = Unknown (coded higher than partially open because this status was often chosen for situations where significant damage was described but there was no mention of whether the road was open or closed)

3 = Blocked

4 = Closed

ii) SBTF-CAP Road Damage by Highway-8Oct2017-2300UTC

1) Text:

This layer summarizes road damage data by highway corridor in order to highlight the Puerto Rican transportation routes most affected by damaged highway infrastructure following passage of Hurricane Maria.

Between October 2nd and October 6th, SBTF and GISCorps volunteers evaluated and geolocated over 1,000 Civil Air Patrol oblique aerial photographs taken after Hurricanes Irma and Maria. These points were intersected with 1-km buffers drawn around priority primary and secondary highways throughout the island. Damage codes for all points within each highway buffer were summed and that sum was divided by total area of the buffer to normalize the data for purposes of comparison.

Damage codes:

0 = Clear or no visible damage

1 = Minor damage

2 = Unknown damage (this was coded higher than minor based on descriptions... often areas were flooded or reports were incomplete but suggested significant damage)

3 = Major

4 = Destroyed

iii) SBTF-CAP Road Status by Highway-8Oct2017-2300UTC

1) Text:

This layer summarizes road status data by highway corridor in order to highlight the Puerto Rican transportation routes most affected by blocked and closed roads following passage of Hurricane Maria.

Between October 2nd and October 6th, SBTF and GISCorps volunteers evaluated and geolocated over 1,000 Civil Air Patrol oblique aerial photographs taken after Hurricanes Irma and Maria. These points were intersected with 1-km buffers drawn around priority primary and secondary highways throughout the island. Status codes for all points within each highway buffer were summed and that sum was divided by total area of the buffer to normalize the data for purposes of comparison.

Status codes:

0 = Open

1 = Partially open

2 = Unknown (coded higher than partially open because this status was often chosen for situations where significant damage was described but there was no mention of whether the road was open or closed)

3 = Blocked

4 = Closed

iv) SBTF-CAP Road Status by Municipality-SBTF-8 Oct 2017-2309UTC

1) Text:

This layer summarizes highway status by municipality in order to highlight the areas of Puerto Rico affected by the highest rates of closed and blocked highways according to an assessment by SBTF and GISCorps volunteers. Between October 2nd and October 6th, the volunteers evaluated and geolocated over 1,000 Civil Air Patrol oblique aerial photographs taken after Hurricanes Irma and Maria.

Status codes for all points within each municipality were summed and that sum was divided by total area of the municipality to normalize the data for purposes of comparison.

Status codes:

0 = Open

1 = Partially open

2 = Unknown (coded higher than partially open because this status was often chosen for situations where significant damage was described but there was no mention of whether the road was open or closed)

3 = Blocked

4 = Closed

v) SBTF-CAP Road Damage by Municipality-SBTF-8 Oct 2017-2309UTC

1) Text:

This layer summarizes highway damage by municipality in order to highlight the areas of Puerto Rico that sustained the most road damage according to an assessment by SBTF and GISCorps volunteers. Between October 2nd and October 6th, the volunteers evaluated and geolocated over 1,000 Civil Air Patrol oblique aerial photographs taken after Hurricanes Irma and Maria.

Damage codes for all points within each municipality were summed and that sum was divided by total area of the municipality to normalize the data for purposes of comparison.

Damage codes:

0 = Clear or no visible damage

1 = Minor damage

2 = Unknown damage (this was coded higher than minor based on descriptions... often areas were flooded or reports were incomplete but suggested significant damage)

3 = Major

4 = Destroyed

vi) SBTF-CAP Highway Status Heat Map-Oct 9 2017-0018UTC

1) Text:

This heat map highlights areas where SBTF and GIS Corps identified clusters of closed or blocked roads in the Civil Air Patrol imagery. Yellow areas indicate a concentration of high road status codes. Status codes were defined as follows:

0 - Road open

1 - Road partially open

2 - Road status unclear

3 - Road blocked

4 - Road closed

vii) SBTF-CAP Highway Damage Heat Map-Oct 9 2017-0018UTC

1) Text:

This heat map highlights areas where SBTF and GIS Corps identified clusters of major road damage in the Civil Air Patrol imagery. Yellow areas indicate a concentration of high damage codes. Damage levels were coded as follows:

0 - Road undamaged

1 - Road sustained minor damage

2 - Road sustained and unknown degree of damage

3 - Road sustained major damage

4 - Road destroyed